Sri Kalyan Ippili

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Summary

Backend & Systems Engineer with a focus on building critical systems from first principles — implementing container runtimes and Merkle-tree tamper-evidence systems from scratch rather than relying on existing libraries. Equally driven by practical problem-solving, building developer tools and automation utilities that address real, firsthand inefficiencies. Experienced across the stack, from low-level systems programming to applied machine learning and developer tooling.

Education

PES University — Bengaluru, India
Bachelor of Technology in Computer Science and Engineering — *2023 – Expected 2027*

Experience

Project Intern — June – July 2026
Centre for Innovation and Entrepreneurship (CIE), PES University — *Bengaluru, India*

- Co-built **SentinelEdge**, a **zero-trust** edge-AI traffic-monitoring platform (3-tier: edge cameras → FastAPI backend → TimescaleDB + Hyperledger Besu) that treats every edge device as untrusted, cryptographically verifies every message (**ECC signatures**, replay protection), and anchors tamper-evident Merkle roots on a private blockchain for offline-verifiable audit/provenance
- Built a **synthetic-city digital twin** of Bengaluru (SUMO + OpenStreetMap, **1,377 signals** across 95 suburbs) generating realistic MQTT traffic calibrated against live TomTom data to drive a production-shaped ingest pipeline
- Designed a continuous **Merkle-tree tamper-evidence** system with a background auditor that automatically detects tampered, deleted, or missing records, anchoring batch roots on a private 4-node **Hyperledger Besu** network
- Fine-tuned a **YOLO** object detection model for vehicle class detection and counting, optimizing for real-world traffic scenarios and edge-device inference
- Built a **k-NN** geographic anomaly detector and sensor-health predictor fused into an incident-vs-compromise correlation layer, distinguishing real traffic incidents from compromised or malfunctioning sensors
- Designed and shipped **Verifiable Provenance Receipts** — a per-reading proof (signature + Merkle inclusion proof + on-chain anchor) independently verifiable offline with zero dependency on the server's code, including a standalone crypto verification implementation

Projects

Nexus — **Open-Source GitOps Platform CLI** | *Python, ArgoCD, Kubernetes, Prometheus/Grafana, Chaos Mesh*

- Built a CLI that takes one config file and one command to a live deployment: installs and wires up ArgoCD for GitOps sync, kube-prometheus-stack for observability, and applies manifests to a Kubernetes cluster, with **478 tests** and **99% core coverage**
- Relies on Kubernetes and ArgoCD reconciliation for **self-healing drift correction**, with optional **Chaos Mesh** fault injection to validate it under failure

Airlock — **Container Runtime from Scratch** | *Go, Linux Namespaces, cgroups v2, BPF/Seccomp*

- Built a functional container runtime from scratch in Go, sandboxing a Linux process with the same primitives Docker is built on: PID/UTS/mount/net namespaces, pivot root, and cgroups v2 memory/CPU limits
- Hand-assembled a **BPF seccomp filter** and native **OCI registry client** with zero dependency on libcontainer, libseccomp, or runc anywhere in the stack

Darwin — **Neuroevolution Engine** | *TypeScript*

- Hand-wrote a representation-blind **neuroevolution engine** with **zero runtime dependencies** where one unchanged core evolves neural networks, strings, numeric vectors, and route permutations across five interchangeable problem environments
- Implemented three search paradigms (competitive co-evolution, generic string/permutation search, network topology evolution) rendered live in-browser as switchable "specimen plates"

Flux — **Cross-Platform File Conversion CLI** | *Go, Pandoc, FFmpeg, ImageMagick*

- Built a cross-platform CLI and Go SDK that routes file conversions across **five engines** (Pandoc, ImageMagick, FFmpeg, a native Go engine for structured data, Python adapters for high-fidelity PDF/DOCX) with smart auto-routing and explicit engine overrides
- Shipped batch/glob conversion, stdin/stdout piping for data workflows, a public Go library package, and a built-in `flux doctor` diagnostics command

Hue — **Real-Time Audio-to-Colour Renderer** | *Swift, GPU Rendering, macOS*

- Built a native macOS app that listens to on-device system audio and renders it as a live **GPU fluid field** of colour, modeled on chromesthesia research (pitch → lightness, emotion → hue, energy → saturation)
- Implemented the colour mapping in a perceptually uniform colour space (**OkLCh**) so equal musical changes produce equal visual changes, rather than an arbitrary beat-chasing palette

The Grand Budapest Hotel (GBH) Suite | *Python, FastAPI, Watchdog, Launchd, macOS*

- Built a silent **macOS automation suite** with file sorting, focus mode, system monitoring, and dotfile backups — all running as LaunchAgents with zero friction
- Developed **11 background "staff" services** that handle real-time file organization, port/log sentinels, CPU runaway enforcement, and git compliance checks
- Engineered a custom backup archiver that strips heavy dependencies (node modules, venv) to reduce archive sizes by **over 90%**, plus live web dashboard for system vitals

Technical Skills

Languages: Python, Go, TypeScript, Swift, SQL
Web & Backend: FastAPI, MQTT, TimescaleDB, Prometheus, Grafana
AI & Machine Learning: YOLO, NLP, Genetic Algorithms / Neuroevolution
Systems & Cloud: Linux Namespaces, cgroups v2, BPF/Seccomp, Kubernetes, ArgoCD, Chaos Mesh
Blockchain & Security: Hyperledger Besu, ECC (secp256k1), Cryptographic Signatures, Merkle Trees, Zero-Trust Architecture
Developer Tools & Cloud: Git, Linux, macOS Launchd, GPU-Accelerated Rendering, SUMO (Traffic Simulation)
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Software Engineering, Cloud Computing

Leadership

Club Head — Appex 2025 – 2026
PES University *Bengaluru, India*

- Directed 'Horcrux 2.0', a **24-hour state-level hackathon**, leading a **120+ member team** from concept to execution on a fixed timeline
- Managed **seven functions** in parallel: technical (track design, judging criteria, event website), sponsorship, design and brand, marketing and campus outreach, logistics and merch, operations and institutional approvals, and event flow
- Organized 'Survivor Island', a DSA-focused coding challenge structured to mirror real technical interview formats

PR and Campaigning Head — Samarpana 2025 – 2026
PES University *Bengaluru, India*

- Led PR and campaigning for Samarpana, PES University's student-run initiative supporting families of Indian martyrs, directing a **40-member team** across digital and on-ground outreach for events with **8,000–10,000 participants** in total
- Grew the official handle by **1,200+ followers** and sustained **30–40k views per post** through a coordinated content and campaign calendar
- Sold out 'Raas-Leher', the flagship Garba night, with **∼3,000 attendees per day**, funding Samarpana's work with martyrs' families every year

Achievements & Certifications

Achievements: 3rd Place, Epoch 2.0 (Intra-college datathon) | Nominated Team, Smart India Hackathon 2024
Certifications: AI & Deep Learning (Pravega, IISc), Machine Learning (LaunchED), Understanding NLP using Python (PESU I/O), Blockchain A-Z & Crypto (Udemy)